

UST'-UDA, Russian Federation

WMO: 305140

Lat: 54.167N

Lon: 103.017E

Elev: 1402

StdP: 13.97

Time Zone: 8.00 (E08)

Period: 82-98

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.5	-29.0	-37.8	0.7	-33.1	-33.3	0.9	-28.6	23.0	5.8	18.5	3.8	2.2	30	0.662

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.7	83.2	65.1	79.6	63.3	75.9	62.1	67.5	77.9	65.7	75.4	63.9	72.6	8.3	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.1	94.2	71.8	62.1	87.9	69.8	60.6	83.1	68.5	32.9	77.6	31.4	75.5	30.0	72.8	74.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.5	17.4	14.9	-39.5	87.1	6.3	4.0	-44.0	90.0	-47.7	92.3	-51.2	94.6	-55.8	97.5		
(5)			-39.9	69.5	5.7	3.4	-44.0	72.0	-47.3	73.9	-50.5	75.8	-54.6	78.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	30.3	-9.8	-4.9	11.5	33.7	47.3	58.9	66.0	61.7	48.4	34.8	15.9	-2.2		
	DBStd	27.65	12.32	11.46	11.28	7.13	7.27	7.03	5.88	5.48	6.57	7.50	11.54	12.00		
	HDD50	8444	1853	1539	1193	490	141	9	0	0	108	471	1022	1618		
	HDD65	12813	2318	1959	1658	939	549	208	60	132	499	936	1472	2083		
	CDD50	1251	0	0	0	1	57	275	497	362	59	0	0	0		
	CDD65	144	0	0	0	0	0	24	91	29	0	0	0	0		
	CDH74	1157	0	0	0	0	33	273	666	183	2	0	0	0		
	CDH80	263	0	0	0	0	3	68	168	24	0	0	0	0		
	WSAvg	6.0	4.2	3.9	5.4	7.2	7.5	6.5	5.9	5.7	6.9	7.1	6.9	4.5		
	PrecAvg	15.6	0.6	0.4	0.4	0.7	1.2	2.2	3.2	3.0	1.7	0.7	0.7	0.7		
(15) Precipitation	PrecMax	19.6	1.7	0.7	1.1	1.7	2.1	3.9	5.9	5.3	3.2	1.4	1.2	1.8		
	PrecMin	11.4	0.2	0.0	0.1	0.4	0.4	1.1	1.6	1.1	0.9	0.2	0.4	0.2		
	PrecStd	2.2	0.3	0.2	0.2	0.4	0.4	0.9	1.0	1.1	0.6	0.3	0.2	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.5	30.3	41.2	61.1	78.2	87.3	88.0	83.4	70.7	56.4	39.5	28.0		
		MCWB	25.2	28.2	35.8	46.7	56.3	63.7	67.2	65.7	59.6	46.4	34.2	25.1		
	2%	DB	17.6	21.5	36.5	54.3	71.4	81.4	84.7	79.2	66.0	51.8	34.8	22.4		
		MCWB	15.8	19.6	31.7	42.3	53.4	61.8	66.3	63.7	55.8	44.3	31.6	20.8		
	5%	DB	11.9	16.3	33.2	49.5	66.5	76.7	81.0	75.3	61.9	48.4	31.9	18.8		
		MCWB	10.6	14.9	29.5	40.5	51.5	59.7	64.9	62.8	53.5	42.2	28.8	17.6		
	10%	DB	7.3	11.9	29.4	45.4	62.0	72.1	77.5	72.3	58.4	45.3	29.6	15.1		
		MCWB	6.3	10.6	26.5	37.3	48.7	58.2	63.7	61.7	51.6	40.5	27.1	14.0		
	0.4%	WB	25.4	28.4	36.5	48.4	57.9	67.6	71.7	68.0	60.2	47.9	34.8	25.3		
		MCDB	26.9	30.5	40.4	57.5	73.9	80.4	83.4	78.8	67.5	54.4	38.1	28.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	16.0	19.8	32.1	44.1	54.9	64.2	68.8	66.0	57.2	44.9	32.0	21.0		
		MCDB	17.5	21.3	35.9	52.6	67.7	76.0	79.6	75.7	63.5	50.6	35.0	22.4		
	5%	WB	10.6	15.0	29.5	40.6	52.2	62.1	66.9	64.5	54.5	43.1	29.4	17.7		
		MCDB	11.7	16.5	32.8	48.7	64.5	72.4	76.9	72.8	60.5	47.8	31.5	18.8		
	10%	WB	6.5	10.8	26.5	37.9	49.4	59.9	65.2	62.7	52.3	40.8	27.2	14.1		
		MCDB	7.4	11.9	29.2	44.4	61.1	69.5	74.3	70.3	57.5	44.9	29.4	15.1		
	5% DB	MDBR	13.1	15.3	19.5	16.6	19.8	17.9	15.7	14.1	13.5	12.1	10.3	12.7		
		MCDBR	15.2	17.3	19.1	23.8	28.6	25.4	22.2	20.1	18.3	16.8	10.5	15.3		
		MCWBR	14.2	15.5	15.4	14.6	14.8	11.3	8.0	8.7	9.8	10.6	8.6	14.1		
		MCWBR	15.2	16.8	18.8	21.8	24.8	19.7	18.0	16.9	16.1	15.4	10.0	14.7		
(35) Mean Daily Temperature Range	5% WB	MCWBR	14.4	15.2	15.2	14.3	13.8	10.4	8.4	8.9	9.7	10.4	8.7	13.7		
	taub		0.285	0.312	0.340	0.383	0.430	0.433	0.433	0.394	0.347	0.330	0.305	0.283		
		taud	2.017	2.004	2.026	2.175	2.108	2.206	2.243	2.336	2.424	2.333	2.126	1.994		
(42) Clear-Sky Solar Irradiance	Ebn at Noon		198	236	259	266	259	257	261	261	235	191	168			
		Edh at Noon	23	33	41	41	46	43	40	35	28	24	21	19		
	RadAvg		338	682	1189	1546	1723	1897	1717	1456	1008	603	318	230		
(45) All-Sky Solar Radiation	RadStd		20	31	48	110	147	120	154	87	90	50	28	23		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page